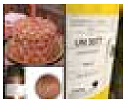
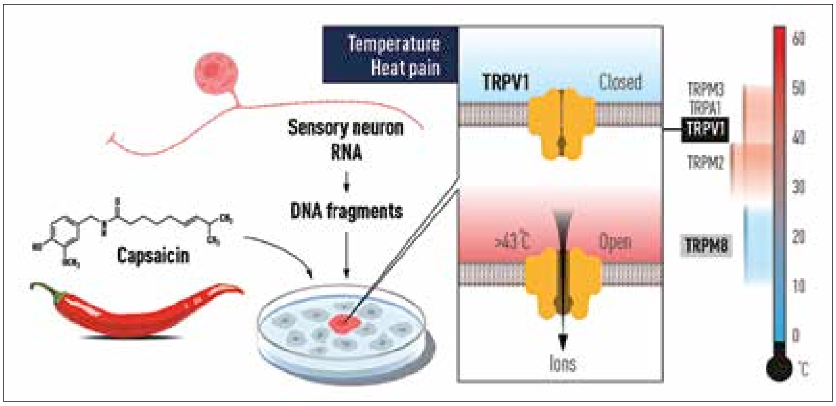


**Poisonous birthday cakes**
The use of "glaze dust" in making birthday cakes is leaving children poisoned by lead and copper. **https://dgit.in/nov21-37**

**Honda makes rockets now**
Honda test fires a small reusable prototype rocket which has been in development since 2019. **https://dgit.in/nov21-38**




Capsaicin and its effect on TRPV1

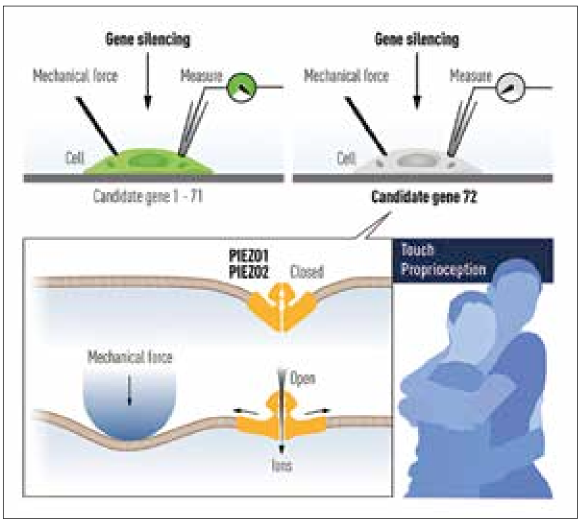
up. Julius and his team, and Ardem Patapoutian, both of whom were working independently, capitalised on this and discovered TRPM8. This receptor worked in a way completely opposite to that of TRPV1. It was activated when the temperature came within a particular range of coldness, and here, in place of capsaicin, menthol was used to activate the receptor.

## NO PRESSURE IS TOO HIGH TO SENSE

It was back in 2002 that Ardem Patapoutian had worked on receptors that reacted to a cold stimulus. The question was how the human body is able to convert mechanical stimuli to electrical impulses. There had been prior research conducted in this area, where scientists had been able to identify mechanical stimuli sensing receptors in bacteria, but it was still a long way to go, given the complexity of the human body.

Ardem Patapoutian, working at Scripps Research in La Jolla, California, USA, along with his team of researchers, made the first major breakthrough in 2010, where it was found that Piezo1 and Piezo2 were components of distinct mechanically activated cation channels. This was accomplished utilising a procedure in which the researchers first identified a cell line that produced a significant amount of electrical signal before inducing a mechanical force stimulation on it by poking it with a micropipette. Following that, 72 candidate genes encoding potential receptors were discovered. Intensive research followed in which, Piezo1 and Piezo2 were identified.

This meant that for the first time in human history, we had a name for a mechanosensitive ion channel, and further research could be done to understand how humans were reacting to, neurologically, the stimuli of pressure. In the following study, whose findings were published in a 2014 research article titled, Piezo2 is the major transducer of mechanical forces for touch




Diagrammatic representation of Piezo1 and Piezo2 and their working

sensation in mice; it was declared that sensory neurons contain a considerably higher number of Piezo2 ion channels as compared to Piezo1 ion channel.

These two ion channels were found in further research to be activated by increased pressure on cell membranes. These findings were used to study the process called proprioception. Proprioception stands for the body's ability to sense movement, action and location. In 2015, a research paper titled, Piezo2 is the principal mechanotransduction channel for proprioception carried information about this process. "We found that the mechanically activated nonselective cation channel Piezo2 was expressed in sensory endings of proprioceptors innervating muscle spindles and Golgi tendon organs in mice," told the team of researchers in the article.

We have different perception mechanisms always working in our bodies, which essentially define the way we interact with the world around us. We have known for years how our sense of hearing works and have made audio products better using that knowledge. Cameras literally work on the same principle as that of the human eye.

So, who knows the different ways in which we get to see this knowledge about perception of temperature and pressure get used for developing or improving consumer products? While that still remains a thing to be seen, there have been significant developments in medical science, like the discovery of a treatment for conditions like chronic pain. These are a testament to the importance of these discoveries.

Not only this but there are also endless uses of these findings of the perception of temperature and pressure that have helped us and are helping us identify and explain various physiological processes and stimuli reactions which have remained a mystery till now. d